Recap: P-series tests - $\sum_{k=1}^{\infty} \frac{1}{k^p}$, if $p>1$ then this series converges. Otherwise it diverges.

Comparison Test - $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ with $0 \leq a_k \leq b_k \ \forall k \in \mathbb{N}$. Then if $\sum_{k=1}^{\infty} b_k$ converges, so does $\sum_{k=1}^{\infty} a_k$.

Alternating series test - $\sum_{k=1}^{\infty} (-1)^k a_k$. If $\lim(a_k)=0$ and (a_k) is monotone decreasing then $\sum_{k=1}^{\infty} (-1)^k a_k$ is convergent.

Ex: i) Consider $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$. Reminder, this is

$$1 - 1/2 + 1/3 - 1/4 + \dots = \ln(2).$$

Define $a_k = 1/k$. Clearly $\lim(1/k) = 0$. Then

$$a_k - a_{k+1} = \frac{1}{k} - \frac{1}{k+1}$$

$$= \frac{1}{k(k+1)}$$

$$\geq 0 \quad \forall k \in \mathbb{N}.$$

Hence, (a_k) is monotone decreasing and by the alternating series test it follows that $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$ is convergent.

ii) Consider the series $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k+1}}$.

Define $a_k = \frac{1}{\sqrt{k+1}}$. Clearly $\lim\left(\frac{1}{\sqrt{k+1}}\right) = 0$

since $\lim(\sqrt{n}) = +\infty$. Then

$$a_k - a_{k+1} = \frac{1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+2}}.$$

Using $\sqrt{k+1} \leq \sqrt{k+2}$, for all $k \in \mathbb{N}$. Then,

$$\frac{1}{\sqrt{k+1}} \geq \frac{1}{\sqrt{k+2}} \Rightarrow \frac{1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+2}} \geq 0.$$

Since (b_n) with $b_n = \sqrt{n}$ is monotone increasing.

Hence (a_k) is monotone decreasing and this by the alternating series test, $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k+1}}$ is convergent.

Q: Can we deduce if $\sum_{k=1}^{\infty} \frac{\sin(k)}{k^2}$ is convergent?

A: Not with techniques so far.

Defn 4.11: A series $\sum_{k=1}^{\infty} a_k$ is said to be absolutely convergent if $\sum_{k=1}^{\infty} |a_k|$ converges.

Theorem 4.12: (Absolute Convergence Test)

If a series $\sum_{k=1}^{\infty} a_k$ converges absolutely, then it converges.

Proof: Suppose $\sum_{k=1}^{\infty} |a_k|$ is convergent. Since,

$$0 \leq a_k + |a_k| \leq 2|a_k|$$

for each $n \in \mathbb{N}$. It follows, by the comparison test and the algebraic limit theorem for series that

$$\sum_{k=1}^{\infty} a_k + |a_k|$$

is convergent. Again, by the algebraic limit theorem for series it follows that

$$\sum_{k=1}^{\infty} (a_k + |a_k|) - |a_k| = \sum_{k=1}^{\infty} a_k$$

is convergent. ☐

Corollary 4.13 : Comparison Test

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be a series with $b_k \geq 0$ for all $k \in \mathbb{N}$. If $0 \leq |a_k| \leq b_k$ and $\sum_{k=1}^{\infty} b_k$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is convergent.

Ex: Consider $\sum_{k=1}^{\infty} \frac{\sin(k)}{k^2}$. Let $a_k = \frac{\sin(k)}{k^2}$.

Clearly, $|a_k| = \left| \frac{\sin(k)}{k^2} \right| \leq \frac{1}{k^2}$.

By the p-series test, absolute convergence test and comparison test it follows that $\sum_{k=1}^{\infty} \frac{\sin(k)}{k^2}$ is convergent.

ii) Consider the series $\sum_{k=1}^{\infty} \frac{k\sin(k)}{k^3+5k^2+3}$. Define

$a_k = \frac{k\sin(k)}{k^3+5k^2+3}$. Then,

$$|a_k| = \left| \frac{k\sin(k)}{k^3+5k^2+3} \right|$$

$$\leq \frac{k}{k^3+5k^2+3}$$

$$\leq \frac{k}{k^3}$$

$$= \frac{1}{k^2}.$$

By the p-series test, the absolute convergence test and comparison test, it follows that $\sum_{k=1}^{\infty} a_k$ converges.

Remark: Note that not all convergent series are absolutely convergent. The obvious example is $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$. This series was shown to be convergent, by the alternating series test. But $\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k}$, this series is divergent.

Defn: Conditional Convergence

A series $\sum_{k=1}^{\infty} a_k$ that does not converge absolutely, but still converges, it is said to be conditionally convergent.

Theorem: Ratio Test

Given a series $\sum_{k=1}^{\infty} a_k$, with $a_k \neq 0$ for all $k \in \mathbb{N}$,

and $\lim\left(\left|\frac{a_{k+1}}{a_k}\right|\right) = L$, it follows that

i) if $L < 1$, the series converges absolutely

ii) if $L > 1$, the series diverges

iii) if $L = 1$, the test is inconclusive.

Proof: i) Suppose $L < 1$. Then there exists a $\alpha \in \mathbb{R}$ such that $0 \leq L < \alpha < 1$. Then there exists $N \in \mathbb{N}$ such that

$$\left|\left|\frac{a_{k+1}}{a_k}\right| - L\right| < \alpha - L$$

$$\Rightarrow \left|\frac{a_{k+1}}{a_k}\right| < \alpha$$

for every $k > N$. It follows that

(By induction)

$$|a_{N+1+k}| < \alpha^k |a_{N+1}|,$$

for all $k \in \mathbb{N}$. Since $\alpha < 1$, the geometric series

$\sum_{k=1}^{\infty} \alpha^k |a_{N+1}|$ is convergent. Hence, by the comparison test, $\sum_{k=N+2}^{\infty} |a_k|$ converges, and thus $\sum_{k=1}^{\infty} |a_k|$ converges.

ii) Suppose $L > 1$. Then there exists $\alpha \in \mathbb{R}$ such that $1 < \alpha < L$. So, there exists $N \in \mathbb{N}$ such that

$$\left| \left| \frac{a_{k+1}}{a_k} \right| - L \right| < L - \alpha$$

$$\Rightarrow \quad \left| \frac{a_{k+1}}{a_k} \right| > \alpha,$$

for all $k > N$. Hence, → (Induction)

$$\alpha^k |a_{N+1}| < |a_{N+1+k}|$$

for all $k \in \mathbb{N}$. As $\alpha > 1$, it follows that $\lim(\alpha^k |a_{N+1}|) = +\infty$. Thus, the sequence $(|a_n|)$ is unbounded and hence $\lim(a_n) \neq 0$. By the divergence test, $\sum_{k=1}^{\infty} a_k$ diverges.

iii) This is a tautology. ☐

Ex: (i) Consider the series $\sum_{k=1}^{\infty} \frac{k}{5^k}$.

Let $a_k = k/5^k$. Then

$$\left| \frac{a_{k+1}}{a_k} \right| = \left| \frac{k+1}{5^{k+1}} \cdot \frac{5^k}{k} \right|$$

$$= \left| \frac{k+1}{5k} \right|$$

$$= \frac{k+1}{5k}$$

$$\longrightarrow \frac{1}{5}.$$

Since $1/5 < 1$, it follows by the ratio test that $\sum_{k=1}^{\infty} \frac{k}{5^k}$ converges.

(ii) Consider the series $\sum_{k=1}^{\infty} \frac{(k!)^2}{(2k)!}$.

Define $a_k = \dfrac{(k!)^2}{(2k)!}$. Then,

$$\left| \frac{a_{k+1}}{a_k} \right| = \frac{((k+1)!)^2}{(2(k+1))!} \cdot \frac{(2k)!}{(k!)^2}$$

$$= \frac{(k+1)^2}{(2k+1)(2k+2)} \to \frac{1}{4}.$$

Since $1/4 < 1$, it follows by the ratio test that $\sum_{k=1}^{\infty} \frac{(k!)^2}{(2k)!}$ converges.

(iii) Consider series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{3^k}{k!}$.

Define $a_k = (-1)^{k+1} \dfrac{3^k}{k!}$. Then,

$$\left| \frac{a_{k+1}}{a_k} \right| = \left| \frac{3^{k+1}}{(k+1)!} \cdot \frac{k!}{3^k} \right|$$

$$= \frac{3}{k+1} \to 0.$$

As $0 < 1$, it follows by the ratio test that the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{3^k}{k!}$ converges.